Akshat Shah

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PROFESSIONAL SUMMARY

Software Engineer and AI/ML Engineer with 5+ years of experience developing scalable backend applications, distributed systems, cloud-native services, and production AI/ML solutions. Proficient in Java, Python, C++, SQL, Spring Boot, PyTorch, TensorFlow, and AWS, with expertise in REST APIs, microservices, machine learning, deep learning, Generative AI, RAG, and multi-agent systems. Experienced in end-to-end software development, model training, fine-tuning, inference optimization, MLOps, and cloud deployment, delivering measurable improvements in system performance, model accuracy, reliability, and operational efficiency.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C++, SQL, JavaScript, TypeScript
Software Engineering & Backend: Data Structures, Algorithms, Object-Oriented Programming, Spring Boot, Spring Security, FastAPI, REST APIs, Microservices, Distributed Systems, System Design, Authentication & Authorization, API Integration, WebSockets, Asynchronous Programming, Database Optimization, Error Handling
Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-Learn, Hugging Face, Transformers, NLP, Sentence Transformers, OpenCV, Computer Vision, Machine Learning, Deep Learning, Feature Engineering, Image Preprocessing, Object Localization, Bounding-Box Prediction, Model Training & Evaluation, Hyperparameter Tuning, Benchmarking
Generative AI, LLMs & Agentic AI: GPT-OSS, Llama, RAG, Embeddings, Semantic Retrieval, Prompt Engineering, Context Engineering, Grounded Generation, SFT, Instruction Tuning, DPO, RLHF, LoRA, QLoRA, PEFT, Multi-Agent Systems, AI Agents, MCP, LangGraph, Tool Calling, Agent Routing, Planning & Reasoning, Agent State Management, Retrieval Evaluation, Grounding Evaluation
Cloud, Databases & DevOps: AWS (EC2, S3, IAM, RDS, Lambda, EKS, ECR, CloudWatch), Azure, Docker, Kubernetes, CI/CD, GitHub Actions, MLflow, OpenTelemetry, Apache Spark, Delta Lake, MySQL, PostgreSQL, MS SQL Server, Git, GitHub, GitLab, Linux, Postman
MLOps, Inference & Optimization: Model Deployment, Model Serving, Experiment Tracking, Model Registry, Drift Monitoring, vLLM, TensorRT-LLM, NVIDIA Triton, Quantization, Batching, Caching, GPU Optimization, Latency Optimization, Throughput Optimization, Model Evaluation
Security, Robotics & Automation: AI Security, Cybersecurity, Adversarial Machine Learning, Backdoor Attack Detection, Prompt Injection Detection, Industrial IoT, Robotic Parameter Optimization, Deepstream, Qt Framework, Real-Time Dashboards, IoT Communication Interfaces, Playwright, W3C WebDriver, Browser Automation, Parallel Execution

PROFESSIONAL EXPERIENCE

Software Engineer Nov 2025 – Jul 2026 | California, USA
Dynamic Active Inc.
- Developed and maintained 20+ scalable REST APIs using Java, Spring Boot, and microservices architecture, supporting 10,000+ daily transactions across business-critical applications.
- Designed reusable backend modules and secure authentication/authorization mechanisms using Spring Security, role-based access controls, and API integrations to improve application security and maintainability.
- Optimized SQL queries, database interactions, and backend processing, reducing API response latency by 30% and improving overall application performance.
- Enhanced software reliability through debugging, automated testing, code reviews, performance tuning, and CI/CD deployments using GitHub Actions, Docker, and AWS.

Research Assistant Mar 2025 – May 2026 | Saint Louis, USA
Saint Louis University
- Conducted research on adversarial vulnerabilities, backdoor attacks, prompt injection, and jailbreak techniques in large language models using Python, PyTorch, and transformer-based architectures.
- Developed semantic analysis and anomaly detection pipelines using Sentence Transformers, sentence embeddings, supervised learning, and unsupervised learning to identify hidden triggers and adversarial patterns.
- Built automated data preprocessing, feature extraction, model training, validation, and evaluation workflows to support repeatable machine learning experiments.
- Collaborated on AI security research, experimental evaluation, technical documentation, and presentations of research findings.

AI/ML Engineer Aug 2022 – Aug 2024 | Vadodara, India
Technoride Solutions Pvt. Ltd.
- Engineered scalable AI/ML and Generative AI pipelines using Python, SQL, Apache Spark, Delta Lake, embeddings, and RAG to transform enterprise data into production-ready context for retrieval and multi-agent applications.
- Developed and optimized PyTorch/Transformer models using Hugging Face, hyperparameter tuning, validation, and evaluation, improving F1 score by 12% and recall by 15%.
- Fine-tuned pretrained LLMs using SFT, instruction tuning, LoRA/QLoRA, PEFT, and DPO/RLHF for domain-specific applications, improving groundedness by 18%, task accuracy by 14%, and agent-routing accuracy by 16%.
- Established reproducible MLOps workflows using MLflow experiment tracking, model registry, version control, Git, CI/CD, and AWS S3/ECR to support model evaluation and production deployment.
- Optimized LLM inference using quantization, batching, caching, vLLM, TensorRT-LLM, NVIDIA Triton, and GPU optimization, reducing p95 latency by 35%, inference costs by 28%, and increasing throughput by 40%.
- Deployed and maintained scalable AI/ML services using FastAPI, Docker, Kubernetes, AWS EKS, IAM, CloudWatch, and OpenTelemetry, implementing model monitoring and drift detection to reduce incident-detection time by 30% and deployment failures by 25%.

Software Developer Mar 2021 – Jun 2022 | Vadodara, India
Wastefull Insights Pvt. Ltd.
- Developed robotic parameter optimization solutions using Deepstream technologies, supporting software functionality for industrial robotics and connected IoT applications.
- Designed and implemented Qt-based industrial IoT dashboards for real-time monitoring, visualization, and analysis of data generated by connected robotic systems.
- Integrated IoT communication interfaces with backend services and cloud-hosted databases, enabling reliable industrial data exchange, processing, and visualization.
- Optimized backend processing and database operations, improving system performance by 25% across industrial software applications.
- Supported software development, debugging, testing, deployment, and production troubleshooting while participating in Agile development and cross-functional collaboration.

PROJECTS

Backdoor Attack Detection in Large Language Models
Python, PyTorch, NLP, Sentence Transformers
- Designed and implemented an AI security framework to identify hidden triggers, semantic anomalies, and backdoor attack patterns in large language model prompts.
- Developed sentence embedding and semantic similarity pipelines using transformer-based language models and Gaussian Mixture Models for unsupervised anomaly detection.
- Processed and evaluated thousands of prompt samples against backdoor attacks, prompt injection, jailbreak attempts, and adversarial scenarios.
- Automated preprocessing, feature extraction, anomaly detection, and evaluation workflows, reducing manual security analysis effort by over 80%.

MCP Server Evaluation & CAPTCHA Automation Framework
Python, Playwright, MCP, LangGraph
- Developed a scalable benchmarking framework capable of evaluating 7,000+ MCP servers through automated health checks, capability validation, tool testing, ranking, and standardized performance assessment.
- Implemented asynchronous MCP client-server communication, tool discovery, execution workflows, and server capability analysis to support distributed automation.
- Built LangGraph-based multi-agent orchestration and Playwright browser automation for parallel server execution and CAPTCHA interaction workflows.
- Automated server evaluation and reporting, reducing manual validation effort by approximately 90% while generating detailed capability and benchmarking reports.

End-to-End Object Localization ML Pipeline
Python, TensorFlow, OpenCV
- Developed an end-to-end computer vision pipeline for image preprocessing, bounding-box prediction, model training, evaluation, and inference using Python, TensorFlow, and OpenCV.
- Trained and evaluated machine learning models on image datasets, applying feature engineering and hyperparameter optimization to improve generalization and prediction performance.
- Developed visualization and evaluation workflows to analyze localization accuracy, bounding-box predictions, and model performance.

ROBOTICS SECURITY RESEARCH

Breaking the Loop: A Systematization of Security in Machine Learning-Driven Robotic Systems.
Akshat Shah (Proposed in IEEE Transactions on Systems, Man and Cybernetics: Systems)
- Conducted a systematic review of adversarial attacks, backdoor attacks, prompt injection, data poisoning, and jailbreak vulnerabilities in machine learning-driven robotic systems.
- Proposed a security taxonomy spanning perception, planning, decision-making, and execution layers, evaluating defense strategies and research directions for trustworthy AI and cyber-physical systems.

EDUCATION

Saint Louis University Aug 2024 – May 2026 | Missouri, USA
Master of Science in Computer Science | GPA: 3.76/4.00

Charotar University of Science and Technology Aug 2020 – Jun 2024 | Gujarat, India
B.Tech in Computer Science | GPA: 4.00/4.00

CERTIFICATES AND COURSES

AWS Certified Cloud Practitioner | AZURE certified AZ-900, AI-900, PL-900 | Google Analytics Certification | Machine Learning Specialization | Cloud Core | Data Analytics | Python for Data Science